Preschoolers can coordinate with each other to communicate about novel referents

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Abstract

Learning language requires learning not only the content of language, but also how to use language to communicate. Iterated reference games provide a window into such skills, requiring rich communication as participants converge on mutually understandable names for initially novel referents. Some early experiments are interpreted as evidence that 4-5-year-old children cannot converge on the mutually understandable names needed to succeed in an iterated reference game. Here, we revisit young children's referential communicative abilities using a simpler, child-friendly version of the itererated reference game paradigm. Across 3 experiments and 82 pairs of children (N=158), we found that 4-5-year-olds could successfully establish reference with each other. Children were 81% accurate, and they often used descriptions similar to their partner's. These findings suggest that children's capacity to construct effective referring expressions in novel contexts emerges earlier than previously thought, consistent with the view that children show early pragmatic competence in supportive contexts.

Keywords: development; iterated reference game; pragmatics; preschoolers

Preschoolers can coordinate with each other to communicate about novel referents

Learning a language requires learning not only the content of that language, but also how to use the language to communicate. One case study for language use is referential communication, the ability to describe a target so an interlocutor can pick it out from a set of possibilities. Adults show sensitivity to both the visual context and their audience during referential communication, calibrating the description they provide to their beliefs about the interlocutor's knowledge state (Brown-Schmidt & Heller, 2018; Galati & Brennan, 2010; Hawkins et al., 2023; Yoon & Brown-Schmidt, 2019).

There is debate in the literature about when children use pragmatics in their interpretation and production of language. On one side of the debate, children seem to learn literal semantics far before they display an understanding of some pragmatic implicatures (Huang & Snedeker, 2009; Noveck, 2001). On the other side, sensitivity to communicative intent is an early emerging skill that develops in parallel with linguistic knowledge and may bootstrap language learning (Bates, 1974; Bohn et al., 2019; Tomasello, 2008).

Iterated reference games provide an important paradigm for studying referential communication. In these games, one player repeatedly describes a set of abstract shapes to a partner so they can identify the target images (Boyce et al., 2024; Clark & Wilkes-Gibbs, 1986; Hawkins et al., 2020; Krauss & Weinheimer, 1964). Over repetition, features of the initial descriptions are conventionalized as each pair comes to agree on a shared understanding of how to label each image.

Children's pragmatic and communicative abilities are commonly assessed in non-iterated contexts interacting with adult experimenters. Iterated reference games played between two children provide a different and richer context to assess multi-turn pragmatic referential abilities and capture a key piece of children's language environment – interaction with peers.

One influential early study in the 1960s claimed that 4-5-year-old preschoolers cannot communicate successfully with each other in an iterated reference game pardigm (Glucksberg et al., 1966). In the past half century, shifts in the interpretive frameworks and methodological considerations of developmental science have raised questions over whether Glucksberg et al. (1966) actually provides compelling negative evidence, given that it is in tension with more with more recent work on the development of referential abilities, which paint a mixed, but overall more capable picture of preschooler's referential communication.

In the rest of the introduction, we review the literature on young children's pragmatic and referential communication abilities, and the literature on child-child communication paradigms. We then focus on the iterated reference game in particular, reviewing the classic Glucksberg and Krauss studies, the questions they raise, and more recent work on iterated reference games with children, before introducing our own study revisiting child-child referential communication in iterated reference games.

Referential communication and pragmatics in young children

Preschooler's show some early-emerging aspects of pragmatic communication, but their demonstration of pragmatic competence is mixed and often masked by task demands.

One aspect of referential communication is understanding how the amount of shared knowledge should effect what types of utterances are appropriate. In producing utterances, children show experimental sensitivity to numerous markers of common ground. While their utterance choices are not equivalent to adults, children produce more informative utterances more when more informative utterances are appropriate (Vasil, 2023).Preschool-aged children adapt the informativeness of their referential expressions based on the visual content available to their interlocutor (Matthews et al., 2006; Nadig & Sedivy, 2002; E. Nilsen & Graham, 2009). Children provide more information when the interlocuter is less knowledgeable (Nayer & Graham, 2006). Children also show sensitivity to other's perspectives when they are the comprehender, using clues to common ground to determine what the correct target is. (Ackerman et al., 1990; San Juan et al., 2015). Children's ability to take partner's perspectives and knowledge states into account is evolving over the preschool years, but overall, by late preschool, children can reason about others' perspectives to communicate more effectively.

Preschool-aged children are also sensitive to the norm of using consistent descriptions in cooperative communication, as they are faster to select a target when it is referred to in a consistent way (Graham et al., 2013; Matthews et al., 2010).

Children are also sensitive as comprehenders to ambiguous utterances, although they have a tendency to pick something even when it is under-specified (Ackerman et al., 1990). By 5 years old, children are sensitive to over- and under-informative utterances, asking for clarification on some under-informative utterances and taking longer to make selections on over-informative utterances (Morisseau et al., 2013).

However, 4-5-year-old children's referential abilities are far from mature. Children struggle to appropriately tailor the specificity of their utterances to the visual context, often resulting in under-informative utterances (Leung et al., 2024; Matthews et al., 2012). In studies where children must pick out one of many referents that vary along combinatorial dimensions (shape, size, color), children often do not mention all the needed dimensions (Bacso & Nilsen, 2017; Deutsch & Pechmann, 1982). When children must describe one of two very similar images, 4 and 5-year-olds sometimes neglect to mention the relevant features, although they do better when playing in an interactive game than when not (Grigoroglou & Papafragou, 2019).

Children are also inconsistent in how they react to feedback after producing an underspecified utterance. Still, children are sensitive to a number of different kinds of feedback. They are especially sensitive to verbal feedback, feedback that asks that child to consider a specific question, and behavioral feedback such as the child receiving an incorrect object. However, while children do provide more information in response to feedback, children often do not produce adequately specified utterances even in response to feedback (Bacso & Nilsen, 2017; Bacso & Nilsen, 2022; Deutsch & Pechmann, 1982; E. S. Nilsen & Mangal, 2012; O'neill & Topolovec, 2001).

Still, children are sensitive to a number of different kinds of feedback. They are especially sensitive to verbal feedback, feedback that asks that child to consider a specific question, and behavioral feedback such as the child receiving an incorrect object.

From an early age, children do not behave in a fully egocentric way, but show signs of tailoring the amount of information they provide in directionally appropriate ways based on the context. At the same time, children often produce utterances that are under-specified, especially when task demands are high.

Kid-kid as a relevant paradigm

Language acquisition has historically focused on how children learn language from direct parental input, but there is increasing acknowledgement that children's language learning occurs across a more diverse set of social interactions (Cristia et al., 2019; Foushee & Srinivasan, 2024). In observational research, there is movement toward studying children's early language in more naturalistic interactional contexts, spanning the range of interlocuters that children interact with across different cultures (Bergey et al., 2025; Casillas, 2023; Girouard-Hallam & Norris, 2024). Cross-culturally, children have rich interactions with other children and create, transmit, and learn from peer cultures (Lew-Levy & Amir, 2024). In contexts where children go to preschool, some of children's linguistic environment is other children in their classes (Chaparro-Moreno et al., 2019; Perry et al., 2018).

Children may have different norms for communication than adults, and so whether children's behavior conforms to adult norms may correspond with whether or not it accomplishes a child's communicative goals. It is important to study how children communicate with each other, both via observation and experiment. There is limited experimental work on young children's communication with each other. In Köymen, Schmerse, et al. (2014), 4- and 6-year-old children identified nameable objects with another child (a similar task to Brennan & Clark, 1996). When the visual context changed so that simpler descriptions would be adequate, 6-year-olds (but not 4-year-olds) were more likely to use consistent referential expressions when they keep talking to the same partner, but switch to simpler terms with a new partner (Köymen, Schmerse, et al., 2014). Other experiments with pairs of children also reveal that children's speech acts when coordinating on an activity are sensitive to what they think is common knowledge with their partner or not (Köymen, Rosenbaum, et al., 2014; Köymen et al., 2016; Short-Meyerson & Abbeduto, 1997). Other child-child communicative studies have looked at iterated learning (Kempe et al., 2019), giving each other directions (Resches & Pérez Pereira, 2007), and asking each other questions (San Martín et al., 2014).

This limited line of research reveals both the availability and sophistication of peer-interactions as a source of early language input and communicative experience. Increased interest in better characterizing the structure of peer-interaction and the role of peer-interaction in language learning warrants further investigation of children's referential abilities in these peer-contexts.

Glucksberg & Krauss's block-stacking studies

Glucksberg and Krauss studied children's referential communication about novel shapes across a number of studies using their "Stack the Blocks" paradigm (Glucksberg et al., 1966; Glucksberg & Krauss, 1967; Krauss & Glucksberg, 1969, 1977).

In Glucksberg et al. (1966), 12 pairs of 4-year-old children were given 6 blocks which depicted different images. The speaker received blocks in a pre-set order from a dispenser, and their job was to stack each block on a peg and tell the other child which block to select and stack on their peg, so the two stacks would match. In the practice phase, children stacked blocks with animals or colored squares on them; first in view of each other, and if that was successful, then separated by an opaque barrier. If children correctly stacked the practice blocks two times with the barrier, they moved on to the critical trials with abstract line drawings on the blocks. 6 pairs of 3-year-olds were tested but did not pass the practice trials. Children received feedback after each round about whether their block stacks matched.

In follow-up studies, Glucksberg and Krauss recruited 7 kindergarten pairs and 9 1st grade pairs from an orphanage to do the block-stacking task (Krauss & Glucksberg, 1969, 1977). They also studied pairs from 3rd through 9th grade and mixed-age pairs with a kindergartener describing to a same-age or older child (Glucksberg & Krauss, 1967; Krauss & Glucksberg, 1969, 1977).

How badly were the children doing? The 4-year-olds never produced an errorless stack of the 6 blocks, and did not show improvement over repetitions (error rates not reported). Kindergarten pairs averaged 2/6 blocks correct per stack (Figure 3 in Krauss and Glucksberg (1969); Figure 4 in Glucksberg and Krauss (1967)), which is numerically above chance, although without knowing the sample sizes and cross-pair variances, we can't say if it is meaningfully different. Adult performance is reported to be generally error-free even in the first repetition, but the 9th graders had less than 50% of their initial stacks completely correct, rising to around 80% correct at the end (Krauss & Glucksberg, 1977).

Glucksberg and Krauss interpret these patterns as evidence for young children's inability to communicate competently. Glucksberg et al. (1966) attributes the 4-year-old's communicative failures to their production of ego-centric descriptions that did not account for the other child's perspective, and Krauss and Glucksberg (1969) interprets the pattern of children's low performance as consistent with early "linguistic competence" followed by much later "communication competence".

Contextualizing classic results & more recent reference games

These studies are cited as evidence that young children cannot form referential pacts with each other, but by the experimental practices of modern developmental science, it's not clear that the Glucksberg and Krauss results should be interpreted as a conclusive failure.

Over time, there have been shifts in how to interpret children's unadultlike behavior in experiments. One framework, expressed in Krauss and Glucksberg (1969), is a "deficit" framing, where children's performance is directly interpreted in contrast to adult. This corresponds with the Piagetian view where young children are expected to be very different than older children and adults.

Another framework is more interested in the developmental trajectory, from the initial emergence of the ability and how it matures across development. This framework focuses more on when and under what conditions there are signs of underlying competence or knowledge, even if the behaviors are inconsistently expressed. This corresponds with theories of core knowledge and naive theory which expect early signals of competence (Gelman, 2024; Gopnik & Wellman, 2012; Wellman & Gelman, 1992).

Along with shifts in theory, methods have also shifted, with a current focus on lowering task demands to pinpoint specific abilities. A common thread among many developmental studies is that children perform better when the cognitive load of the task is reduced. For instance, in appearance-versus-reality tasks, Sapp et al. (2000) and Rice et al. (1997) revealed children had earlier understanding of the difference between appearance and reality by using tasks with lower demands on working memory and executive functioning. Meta-linguistic judgments are harder for children than active tasks that probe the same pragmatic constructs (Ackerman et al., 1990; Porrini & Surian, 2023). In reference tasks, children perform better when there are fewer possible referents (Abbot-Smith et al., 2015; San Juan et al., 2015).

Revisiting the Krauss and Glucksberg results under this framework, the interesting question is no longer are 5 year olds worse at things than adults, but instead trying to understand whether and to what extent the rudiments of competencies are present at age 5. The performance of the youngest children is not strong evidence of complete failure, but rather weak evidence. It is unclear if the children in the Krauss and Glucksberg studies are statistically above chance, but it is clear that there were substantial task demands. A task that differentiates 9th graders from adults and includes 6 items at a time is likely not the most sensitive measure of the communicative abilities of 4 and 5-year-olds.

Since Glucksberg et al. (1966), few studies have revisited the question of whether children can successfully communicate in an iterated reference game. In one study, 8-10-year-olds exhibited adult-like patterns of increasing accuracy, increasing speed, and shorter descriptions across repetitions, but children's accuracy was still far below adult performance and highly variable between dyads (Branigan et al., 2016). In an iterated reference game where 4- and 6- year-olds communicated to each other via gestures, children were successful at communicating a repertoire of 5 image options with each other (Bohn et al., 2019). Children's comprehension increased over repetitions, and when children switched roles, they often re-used their partner's gestures. The 6-year-olds showed greater within-dyad similarity in gestures than the 4-year-olds, indicating that children's abilities to form conventional pacts is increasing in this age range.

Leung et al. (2024) ran an iterated reference game where 4-8-year-old children used natural language to communicate with their parents. In this simple, child-friendly tablet-based task, even 4-year-olds had an initial accuracy above 80% on two-way choice, which rose to above 90% in later repetitions (Leung et al., 2024). Leung et al. (2024) interpret their results in contrast with the Glucksberg et al. (1966) results as a sign that parental scaffolding is what enables children's successful matching. While parents certainly scaffold their interactions with children, it is not clear whether the children's better performance is due only to the parental communication partners or the reduced task demands.

Current study

In our current study, we adapt the paradigm used in Leung et al. (2024) to re-examine young children's ability to establish effective referring expressions with each other. Compared to Glucksberg et al. (1966), we have reduced task demands in that children do not need to stack blocks, instead the describer only needs to say what the target is, and the matcher to select it. We had a total pool of only 4 options in each experiment and only 2 options each trial, rather than the 6 images Glucksberg et al. (1966) used. We also tested many more children: 82 pairs of 4-5 year old children across 3 studies, rather than the 12 4-year-old pairs and 7 kindergarten pairs of Krauss and Glucksberg.

Across our three studies including a total of 82 pairs of 4-5-year-old preschool children, we found that preschool-aged children were successful in an iterated reference game, suggesting that children's capacity to construct effective referring expressions in novel

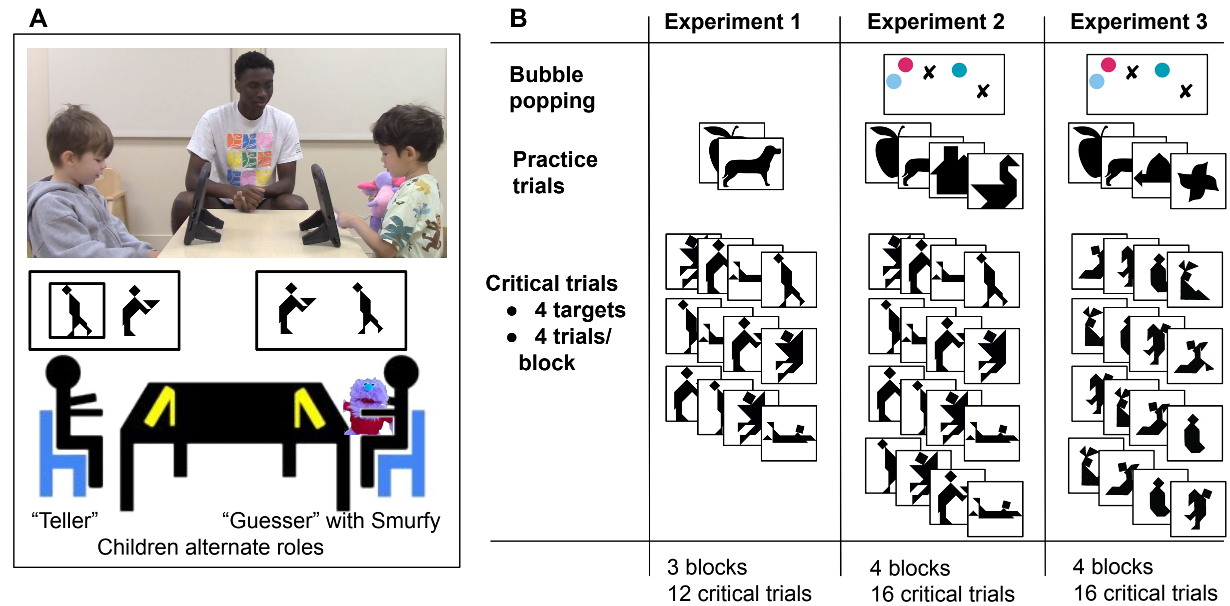

Figure 1. Experimental setup and procedure. Panel A shows the experimental setup. Panel B shows the procedures for each experiment; within critical blocks targets were ordered randomly.

contexts emerges earlier than previously claimed.

Experiment 1

Methods

Our goal for Experiment 1 was to test young children's ability to coordinate to produce descriptions of abstract shapes that their partner could understand. Young children can be very sensitive to task demands and cognitive load (Carruthers, 2013; Keen, 2003; Turan-Küçük & Kibbe, 2024), so we adapted the experimental framework from Leung et al. (2024), and further simplified it by reducing the total pool of targets and the number of trials. This experiment was pre-registered at osf.io/kcv8j.

Participants. 4 and 5-year-old children were recruited from a university preschool laboratory preschool located in the Bay Area, California. This nursery school is a Montessori-like preschool that emphasizes a child-directed, play-based curriculum. The children who attend this preschool are mostly from high socio-economic backgrounds. This research was approved by the [university name] IRB, and the children's parents had given written informed consent for their children to participate in research at the preschool.

Children played with another child from the same class. Experiment 1 was conducted between June and August 2023. We targeted 20 pairs of children completing enough critical trials, based on what felt feasible to collect in a summer and common developmental sample sizes. Pairs of children were included in analyses if they completed at least 8 of the 12 critical trials. We had 19 games that completed 12 critical trials, and 1 game that completed 11 critical trials. Of the 40 children, 21 were girls, and the median age was 4;9, with a range of 4;0 to 5;10. In terms of parent-reported ethnicity, 12 children were White, 9 were Asian, 8 were mixed race, 3 were Black, 4 were Hispanic, and 4 did not have race specified. An additional 7 games were excluded for being pilot runs or not completing enough trials.

Materials. For the target stimuli, we used four of the ten tangram images from Leung et al. (2024), chosen based on visual dissimilarity (Figure 1B). We coded the matching game using Empirica (Almaatouq et al., 2021), hosted it on a server, and then accessed the game on tablets that were locked in a kiosk mode so children could not navigate away from the game.

Procedure. Once a pair of children agreed to play the game, a research assistant took them to a quiet testing room. The RA introduced the children to a stuffed animal "Smurfy" who wanted to play a matching game, and then explained how the game would work. Children were informed that they could ask to stop playing and go back to the classroom at anytime. Children sat across a table from each other, each with a tablet in front of them (Figure 1A). On each trial, one child was the "teller" and saw a black box around one of two images on their screen and was asked to "tell Smurfy what they see" in the black box. The "guesser" saw the same two images in a randomized order and tried to select the described image to help Smurfy make a match. When the guesser selected an image, both children received feedback in form of a smiley or frowny face and an excited or disappointed sound. After each trial, children switched roles. Children passed Smurfy back and forth to keep track of whether they were the "guesser" or "teller" on a given trial.

Children completed two warm-up trials with black and white images of familiar shapes, followed by 3 blocks of the 4 target images (Figure 1B). Targets were randomly paired with another of the critical images as the foil.

The experimenters running the game did not volunteer descriptions, but they did scaffold the interaction, prompting children to describe the images, and sometimes repeating children's statements (especially when utterances were inaudible or the child did not respond immediately; this aspect of the procedure was modified in Experiment 2 and 3). The entire interaction was video-recorded.

Comparison to methods in prior work. In our current study, we adapt the paradigm used in Leung et al. (2024) to re-examine young children's ability to establish effective referring expressions with each other. Our current study is of 4 and 5 year old preschoolers, who are most similar in age to the 4 year old preschoolers and kindergarteners in Glucksberg et al. (1966) and Krauss and Glucksberg (1969), and also comparable in age to the 4-year old sample in Leung et al. (2024).

The key differences from the Leung et al. (2024) study are that we only did 3 blocks of critical trials (compared to their 4), and that we only used 4 target images (a subset of their 10). Combined, this meant that our experiment was much shorter with 12 critical trials rather than 40. Leung et al. (2024) did not have feedback on the correctness of a match, and we did. Rather than have a parent who played the game and also kept the child on task, we had an RA who was not participating in the game but did prompt children as needed to keep them on task. We introduced the framing of "helping smurfy make matches" to help children understand the game and keep track of whose turn it was to guess and to tell.

The methodological details of Glucksberg et al. (1966) are less clear. We do not know what support (if any) experimenters gave children to keep them on task. The cover task was different, as Glucksberg et al. (1966) had children stacking blocks and we had children pressing images on a tablet. We had a total pool of only 4 tangram images in each

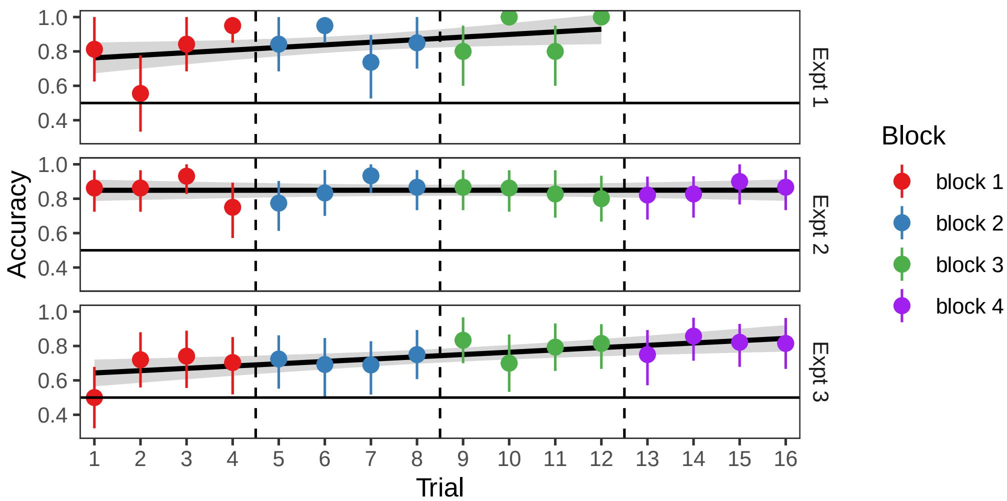

Figure 2. Children's accuracy at selecting the correct target over time. Error bars are bootstrapped 95% CIs with a linear trend line overlaid.

experiment and only 2 options each trial, rather than the 6 line drawing options Glucksberg et al. (1966) used. Children in our study alternated roles, rather than having a fixed role, and got feedback after every match, rather than every 6 blocks. The overall experiment was much shorter, intended to fit within 15 minutes, rather than the 50 minute-limit (Krauss & Glucksberg, 1969).

Data processing. Children's selections and the time to selection were recorded from the experiment software. Children's descriptions were automatically transcribed from the video using Whisper (Radford et al., 2022) for the first pass and then hand-corrected by experimenters. Transcripts were hand-annotated for when each trial started, who said each line, and what referential descriptions were used. We excluded 9 trials where the "teller" did not produce a description, where all description was unintelligible and impossible to transcribe, or where the "guesser" did not make a selection. After exclusions, we had 230 trials remaining.

Statistical analyses were run in brms (Bürkner, 2018) with weakly informative priors. We report estimates and 95% credible intervals. The experimental set-up, analysis code, de-identified transcripts, and performance data for all experiments is available at osf.io/7d3ne.

Results

Accuracy and speed. Our primary question was whether children could accurately communicate the intended target, as prior work is often interpreted as indicating the children at kindergarten age cannot communicate about abstract shapes successfully (Glucksberg et al., 1966). To test for changes in accuracy over time, we fit a Bayesian mixed effects logistic regression predicting accuracy.[1] Children's accuracy was above chance (Odds

[1] correct∼ trial + (trial|game) + (trial|target) Priors were normal(0,1) for beta and standard deviation of random effects and lkj(1) for correlations between random intercepts and random slopes. In this and other models, we deviate from the pre-registration by adding random by-image slopes to better match the

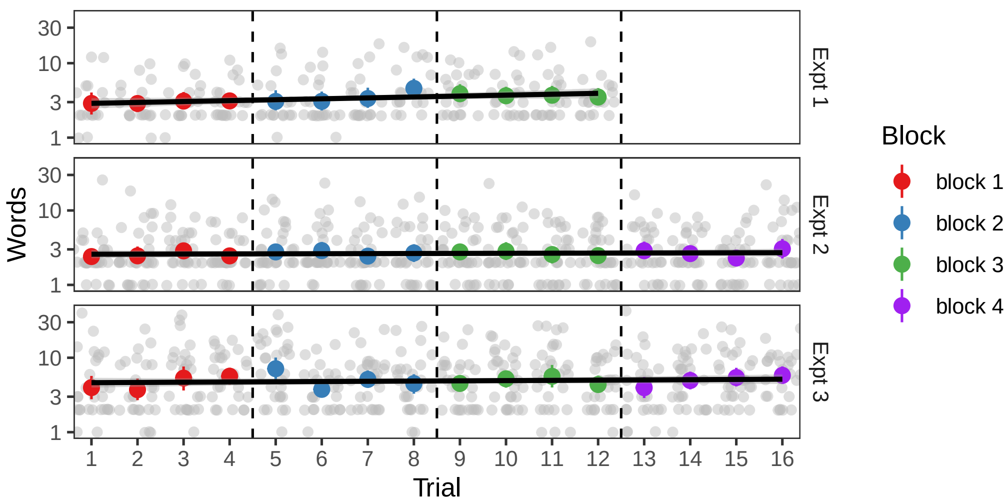

Figure 3. Length of description produced by the teller each trial, shown on a log scale. Grey dots are individual data points, colored dots are per trial means with bootstrapped 95% CIs.

Ratio: 3.11 [1.13, 9.00]) with a possible increase in accuracy over the game (OR of one trial later: 1.17 [0.97, 1.44], Figure 2). This level of accuracy is generally in-line with accuracies from 4-year-olds playing with their parents in Leung et al. (2024) and indicates that children can understand and succeed at the task.

As another measure of children's performance, we looked at how long children spent on each trial. We ran a Bayesian mixed effects linear regression predicting the time to selection in seconds.[2] The first critical trial averaged 27.64 [15.98, 38.54] seconds, and children tended to speed up over time, albeit with wide variability (-1.13 [-2.65, 0.42] seconds / trial). Children were able to achieve the same accuracy in less time, suggesting that they were becoming more efficient at completing the task.

Description length. In iterated reference games with adults, description lengths usually shorten over repeated references (Boyce et al., 2024; Clark & Wilkes-Gibbs, 1986; Hawkins et al., 2020). We were curious if children's descriptions would display the same trend, so we ran a Bayesian mixed effects linear regression predicting the number of words in the description the "teller" produced.[3] On the first critical trial, descriptions averaged 3.63 [2.04, 5.25] words, and description length was relatively stable over time (change of 0.12 [-0.22, 0.56] words per trial, Figure 3). Thus, children's increasing speed was not from shorter utterances, but instead from some combination of improved task understanding, faster utterance planning, and faster decisions of what to select.

Some examples to illustrate the variety of effective descriptions children employed are

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experimental design.

[2] time.sec~ trial + (trial|game) + (trial|target) Priors were normal(60,100) for the intercept, normal(0,20) for betas and for standard deviations of the random effects, and lkj(1) for correlation between random intercepts and random slopes.

[3] words~ trial + (trial|game) + (trial|target). Priors were normal(5,10) for the intercept, normal(0,5) for beta and standard deviation of the random effects, and lkj(1) for correlation of the random intercepts and random slopes.

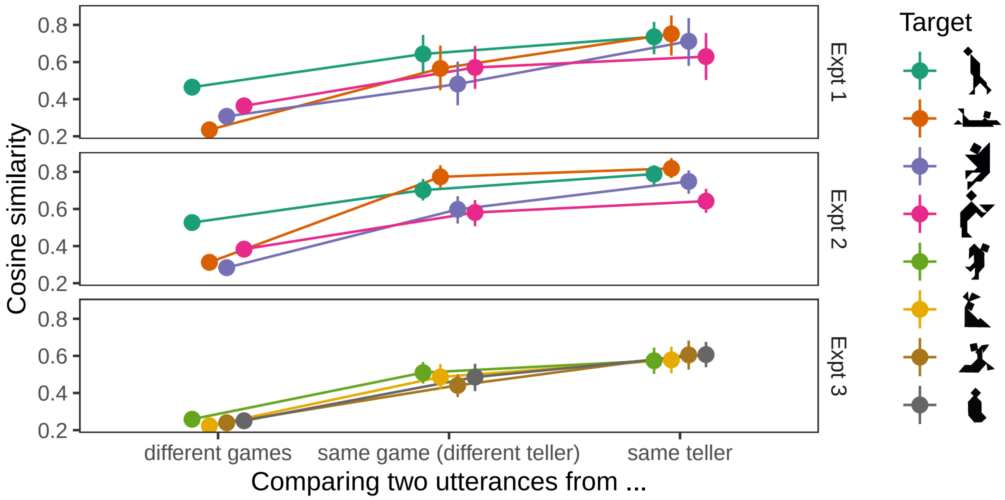


Figure 4. Semantic similarity between pairs of descriptions from different sources. Dots are means and lines are bootstrapped 95% CIs.

shown in Table 1.

Convergence. While description length is often used as a proxy for measuring convention formation, it does not capture semantic overlap between utterances. Boyce et al. (2024) introduced a more sensitive measure of semantic convergence that compares the content of utterances using word embeddings to trace how similarities within and across games change over time. To measure convergence of utterances, they compared how the similarities between utterances from a given block and the final block (conventionalized) utterances increased over the course of the game. With only 3 blocks of descriptions, we do not expect semantic similarity for descriptions of a given target to show any meaningful change over time. However, we can test for a more coarse measure of sensitivity to partner: whether children's utterances are more like the utterances of their partner than like utterances from children in other games. If children are fully ego-centric (as suggested by Glucksberg et al., 1966), their choices of descriptions would be independent from their partners, and equally similar (or dissimilar) to descriptions from their partner and from other peers.

Following the methods of Boyce et al. (2024), we embedded each description in a semantic vector space using S-BERT (Reimers & Gurevych, 2019), and then used the cosine between embeddings as a measure of semantic similarity.

We compared the semantic similarities between descriptions of the same target based on who produced the description (Figure 4). We used a Bayesian mixed effects linear regression to predict similarity.[4] Utterances were more similar if they came from the same partnership (0.198 [0.114, 0.266]). Utterances were slightly more similar still if they came from the same person (0.127 [0.043, 0.201]), which is expected since children are likely to be fairly consistent with themselves. However, children used descriptions that were much more

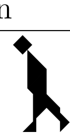
[4] sim∼ same_game + same_speaker + (same_game + same_speaker|target) This and all other models using cosine similarity as the dependent measure used the priors of normal(0.5, 0.2) for the intercept, and normal(0, 0.1) for beta, and normal(0, 0.05) for the standard deviation of random effects.

similar to their partner's than to other children's (Figure 4), indicating sensitivity to their partner's expressions.

Discussion

In Experiment 1, we adapted the paradigm of Leung et al. (2024) for pairs of children, taking an already simple set-up and making it shorter. Our goal was to see if young children were at all able to provide adequate descriptions, so children received a lot of scaffolding around the experimental interaction. In this supportive environment, children provided short descriptions that allowed their partner to select the correct target most of the time. The scaffolding provided by experimenters sometimes included echoing children's descriptions, which could potentially influence children's responses. In Experiment 2, we repeated the same paradigm, with a tighter experimental script and a larger sample size.

Table 1
Example descriptions children successfully used to identify different target images in Experiments 2 and 3.

• vampire • a person flying • a person • a kite • a triangle with a head on it with feet • somebody skydiving, not in the airplane 	• a people • a horse • stomping • a person that's walking and has a square, like on his shoulder • a diamond on the back, this leg is one leg on the ground, one leg up in the air 
• airplane • alligator • a person fell down • a boat • a person that's in a race car that has one triangle and two triangles 	• a bunny • a kangaroo • a mountain with some triangles • two triangles on the top • a person looking backwards with a dress and a bow 
• person • a person holding a sandwich • a people carrying a box of dirt • a monster • someone holding a plate and giving it to a restaurant and has watermelon 	• a shoe • a walrus • something looking down • a pterodactyl with a flag on its back • a diamond on the top, and a triangle on the bottom 
• person • a person walking • a person looking down • a people, but it doesn't have any arms 	• a bottle • a diamond on the top • a weird looking banana • candy 

Experiment 2

Methods

As Experiment 2 was very similar to Experiment 1, we focus on the differences from Experiment 1. Experiment 2 was pre-registered at osf.io/y2dax.

The biggest change between the experiments was increasing the number of repetitions of target stimuli from 3 to 4 (from 12 to 16 trials). The greater number of trials in Experiment 2 made it possible to look for changes over time that could be indicative of convergence to shared descriptions within a game and divergence between games.

Participants. Experiment 2 was run between March and August of 2024, at the same preschool as Experiment 1. We aimed to collect data from 30 pairs of children, to increase the precision of our effect estimates compared to experiment 1. We did not intend to resample children who had already participated in Experiment 1; however, after the end of data collection, we noticed that 5 children included in Experiment 2 had previously been part of Experiment 1.[5] Pairs of children were included in analyses if they completed at least 8 of the 16 critical trials. 30 pairs of children completed all 16 critical trials, and 1 pair of children completed 10 critical trials. Our target age range was 4 and 5-year-olds, but one older 3-year-old was unintentionally included. Of the 62 children, 30 were girls, and the children had a median age of 4;8, and a range of 3;9-5;9. In terms of parent-reported ethnicity, 16 children were Asian, 16 were mixed race, 10 were White, 6 were Hispanic, 1 was Black, and 13 did not have race specified. An additional 6 games were excluded for being pilot runs or not completing enough trials.

Materials. The same 4 critical images were used as in Experiment 1. In response to some children struggling with the abrupt switch from familiar to non-nameable shapes, we introduced more practice trials for Experiment 2. We used a total of 4 practice trials to provide a gradient from familiar shapes to less recognizable, blockier shapes (Figure 1B).

Procedure. The procedure was much the same as Experiment 1(Figure 1B). We added an initial "bubble popping" exercise to give children practice tapping the tablet appropriately (this was an issue for some children in Experiment 1). The experimental script was fully written out and memorized by experimenters so children all received the same instructions. We wrote up contingency statements that the experimenter could use to prompt children who were not giving descriptions or making selections. Experimenters helped with game mechanics such as whose turn it was to tell and who should press the screen, but avoided contributing or repeating any content about the images or the descriptions.

Data processing. Data were processed in the same way as Experiment 1. After excluding trials where children did not give a description or where the experimenter echoed a child's description (15 trials), we had 471 trials total.

Results

Accuracy and speed. In Experiment 2, children's accuracy was above chance (Odds Ratio: 5.87 [2.61, 13.38]) and relatively stable over time (OR of one trial later: 1.01

[5] One Experiment 2 game had two children who had been in Experiment 1 pilots, two games each had one child previously included in Experiment 1, and one game had a child who had started an Experiment 1 game that was excluded for too few trials.

[0.88, 1.15], Figure 2). The first critical trial averaged 20.66 [9.65, 32.42] seconds, and children's speed increased over repetitions, with wide uncertainty (-0.69 [-1.70, 0.31] seconds / trial). Taken together, we find more evidence that children can successfully communicate with each other about these abstract shapes, and do so with increasing efficiency.

Description length. The average length of descriptions on the first trial was 3.28 [1.66, 4.82] words and description length was relatively stable over time (change of 0.02 [-0.08, 0.13] words / trial, Figure 3). This finding is comparable to Experiment 1, again finding that children produce short utterances without much change in length over time.

Convergence. As a coarse measure of partner-sensitivity, we repeated the semantic analysis from Experiment 1. Utterances were more similar if they came from the same partnership (0.225 [0.121, 0.314]) and were slightly more similar if they came from the same person (0.084 [0.031, 0.134]).

As Experiment 2 had 4 blocks, we examined whether descriptions were converging semantically toward the final description. We compared the utterances from the first three blocks to the descriptions in the last block using a Bayesian mixed effects linear regression predicting similarity.[6] Over the first three blocks, descriptions became increasingly similar to the last block description (0.076 [0.033, 0.118]). Descriptions from the same child would be expected to be more similar than descriptions from the partner as a sign of internal consistency; descriptions from the same child were only marginally more similar, with wide uncertainty (0.048 [-0.026, 0.118]). Although over time descriptions did get more similar to the last block utterance, the semantic distance between adjacent block utterances did not clearly increase over repetition (0.030 [-0.020, 0.078]).

Partnerships may also diverge from one another as groups focus on distinct aspects of the image. We tested whether descriptions in different games diverged over time using a Bayesian mixed effects linear regression.[7] As the games progressed, descriptions to the same target from different games did not change substantially in similarity (-0.011 [-0.039, 0.019]).

Discussion

In Experiment 2, we repeated the matching game from Experiment 1 with a tighter experimental script, a larger sample size, and more trials per game. We again found that children produced short descriptions that enabled their partners' to identify the target most of the time. Children's descriptions showed semantic convergence toward shared conventions for referring to each target within games.

Experiment 3

The first two experiments demonstrated children's ability to communicate referentially, but were limited in their generality by the use of a specific set of 4 fairly humanoid tangram figures. To explore the generality of the results to a different set of potentially harder tangram stimuli, we conducted Experiment 3.

[6] sim∼ earlier + same_speaker + (earlier + same_speaker|game1) + (earlier + same_speaker|target)

[7] sim∼ block1 + (block1|target)

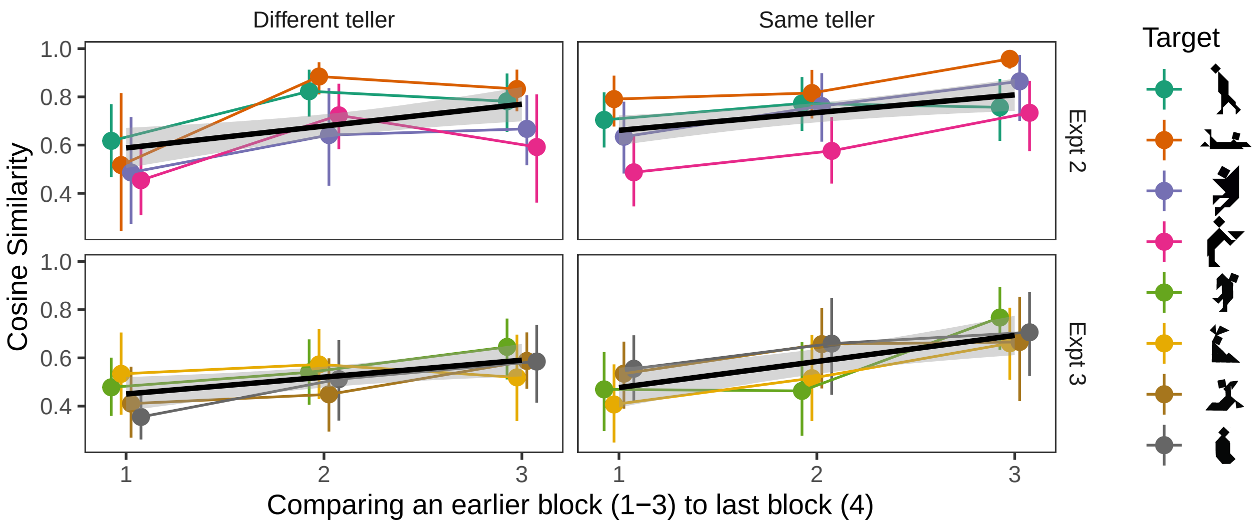


Figure 5. Semantic similarity between descriptions from earlier blocks (1-3) and the last block in Experiments 2 and 3. Heavy dots are means with bootstrapped 95% CIs.

Methods

As Experiment 3 was very similar to Experiment 2, we focus on the differences from Experiment 2. Experiment 3 was pre-registered at osf.io/tk7sc.

The procedure for Experiment 3 matched Experiment 2, and the only change was in the stimuli used.

Participants. Experiment 3 was run between December 2024 and January 2026, at the same preschool as Experiments 1 and 2. We targeted collecting 30 pairs of children, to match the sample size of Experiment 2. Because the stimuli changed, we allowed children who had participated in a prior experiment to participate again. 6 children included in experiment 3 had previously been part of experiment 2 (4 of whom were included in experiment 2 data, 2 of whom had been in experiment 2 pilots). 30 pairs of children completed all 16 critical trials, and 1 other pair completed 11 trials. Of the 62 children, 33 were girls, and the children had a median age of 4;9, and a range of 4;0-5;10. In terms of parent-reported ethnicity, 22 children were White, 14 were mixed race, 13 were Asian, 8 were Hispanic, 1 was Black, and 4 did not have race specified. An additional 17 games were excluded for being pilot runs, including repeat participants, not completing enough trials, or including a 3-year-old participant.

Materials. A different set of critical images were used (see Figure 1B). These were chosen from a set of commonly used tangram images (Boyce et al., 2024; Clark & Wilkes-Gibbs, 1986; Hawkins et al., 2020), to be fairly distinct from each other. We also switched the stimuli for the 3rd and 4th practice trials to be more abstract, so children would not be surprised by the critical stimuli not having established, canonical names.

Procedure. The procedure was the same as for Experiment 2.

Data processing. Data were processed in the same way as Experiment 1. After excluding trials where children did not give a description or where the experimenter echoed a child’s description (30 trials), we had 446 trials total.

Results

Accuracy and speed. In Experiment 3, children's initial accuracy was slightly above chance, but with wide uncertainty (Odds Ratio: 1.76 [0.87, 3.62]). Children's accuracy tended to increase slightly over time, again with some uncertainty (OR of one trial later: 1.08 [0.98, 1.19], Figure 2). Children's overall accuracy was lower in Experiment 3 (74%) than in the prior experiments (85%), potentially indicating that the stimuli were more difficult.

The first critical trial averaged 27.99 [21.14, 34.78] seconds, and children got faster over time (-0.97 [-1.52, -0.42] seconds / trial). Overall, we find evidence that children's ability to communicate with increasing efficiency generalizes to different abstract images.

Description length. The average length of descriptions on the first trial was 7.06 [4.82, 9.17] words and description length was relatively stable over time (change of -0.04 [-0.35, 0.25] words / trial, Figure 3). Descriptions in experiment 3 averaged 7 words, compared to an average of 3-4 words in the prior experiments. Again, the difference may be due to the increased difficulty of the stimuli.

Convergence. We repeated the semantic analyses from Experiment 2. We again found that utterances were more similar if they came from the same partnership (0.233 [0.195, 0.263]) and if they came from the same person (0.107 [0.058, 0.148]).

Over the first three blocks, descriptions became increasingly similar to the last block description (0.080 [0.035, 0.123]). Descriptions from the same partner were slighly more similar to each other, with some uncertainty (0.068 [-0.008, 0.140]) Over time, the semantic similarity between adjacent block utterances increased (0.066 [0.017, 0.112]).

Unlike the patterns found with adults, we found that the similarity between utterances in different games was relatively stable over time (0.007 [-0.008, 0.025]). While children show some of the same patterns as adults on these stimuli, such as sensitivity to their partners descriptions, and some convergence towards conventions, there was not increasing group-specificity of descriptions over time.

Discussion

In Experiment 3, we tested whether preschool children's ability to coordinate on mutually understandable labels to identify targets in a repeated matching game would generalize to a different, and perhaps harder, set of abstract stimuli. In line with the prior experiments, children used relatively short descriptions, picked the correct targets most of the time, and showed signs of converging with their partner on names for each image. Possibly because the images were more difficult, children used longer descriptions and had lower accuracy than in the first two experiments.

Pooled analysis

We conducted an exploratory analysis over the pooled dataset first in order to understand the consistency of specific results across experiments and secondly to explore potential moderators of children's behavior across the larger dataset.

Consistency of results

Because of variability between pairs of children and between different target images, there is variability in our numerical estimates of results across different experiments. In order to maximize power and obtain overall estimates of effects, we re-ran the statistical models, pooling across the experiments. In addition to random slopes and intercepts for each pair and each image, we add random slopes and intercepts by experiment as well.

Accuracy and speed. Pooling across the three experiments, children's accuracy was above chance at the start of the game (OR: 3.04 [1.01, 8.40]), but it is unclear whether children's accuracy increased over the course of the game (OR of one trial later: 1.06 [0.87, 1.29]). Across all the experiments, it is unclear whether children sped up over the course of the game (effect of one trial later: -0.90 [-2.18, 1.20] seconds).

Description length. Descriptions produced by tellers averaged 4.74 [0.57, 8.94] words on the first trial, and description length was relatively stable over time (change of 0.03 [-0.27, 0.53] words per trial).

Convergence. Utterances were more similar if they came from the same partnership (increase in cosine similarity: 0.221 [0.138, 0.287]) and were more similar still if they came from the same person within the partnership (0.101 [0.049, 0.150]).

For the other semantic analyses, we pooled across experiments 2 and 3 which had 4 blocks. In the combined model, we found that the similarity the last block utterance increased across blocks (0.074 [0.008, 0.120]) and was marginally higher, albeit with some uncertainty, when the same child produced both utterances (0.061 [-0.018, 0.132]). In a combined model, similarity to the next block utterance also increased across blocks (0.047 [-0.008, 0.094]) and was higher when the same child produced both utterances (0.136 [0.064, 0.197]). Across the two experiments, the similarity of utterances from different pairs to the same stimulus remained fairly stable across blocks (-0.005 [-0.031, 0.025]).

Different pairs of children used different descriptions for the target images, and within a partnership, children's descriptions show convergence toward a shared concept. However, this convergence was not coupled with increasing dissimilarity to the descriptions of other pairs. The overall pattern of results is a partial match for the patterns reported in reference games between adults (Boyce et al., 2024).

Sensitivity to prior communicative success

One aspect of mature cooperative communication is tailoring descriptions to the knowledge state of the interlocuter. If an interlocutor is confused or misunderstands a description, adults try a different description or add more details to help their interlocuter. In iterated reference games, adults shorten descriptions more when the prior description was successful compared to when it was not (Hawkins et al., 2020).

If children are sensitive to their partner's perspective, then whether a description was successful should influence whether the same description, or a variant, will be used in future rounds. To test whether accuracy is predictive of similarity to future descriptions, we ran a post-hoc Bayesian linear model predicting similarity to the next block description in terms of accuracy.[8] Descriptions that elicited a correct response were more similar to the next block

[8] sim$\sim$ earlier $\times$ correct + same_speaker $\times$ correct + (earlier $\times$ correct + same_speaker $\times$ correct|game1) +

description (0.102 [0.003, 0.198]) with no substantial interaction with block number or whether both descriptions came from the same teller. This pattern of results is consistent with the expectation that children are more likely to stick to their own description or repeat the other child's description if it was previously successful.

Age and gender as moderators of referential success

We consider the effects of participant age and gender as an exploratory measure. Children were not matched with their partner based on age and gender, so different ages (within the 4-5 year old target age range) sometimes played together, and children sometimes played with same-gender and sometimes with different-gender partners.

Accuracy. To test for possible effects of children's age and gender on matcher accuracy, we ran a mixed effects logistic regression, where we considered possible effects of the age of each child (centered at mean age of all the children who participated) and the gender of each child on the accuracy of their selections[9]. Accuracy was higher for older children (OR for one month older teller: 1.04 [1.01, 1.07], OR for one month older guesser: 1.04 [1.01, 1.07]). Accuracy was also higher when the guesser was a girl (OR: 1.52 [1.08, 2.17]), but was not credibly higher when the teller was a girl (OR: 1.34 [0.96, 1.89]).

Description length. We also tested whether children's age or gender affected how many words they produced in their descriptions. We ran a mixed effects linear regression, where we considered the ages and genders of both children as possible predictors of description length[10]. Older children tended to produce longer descriptions (effect of one month older 0.20 [0.11, 0.29]), but there was no effect of age of guesser (-0.03 [-0.12, 0.05]). For gender, there was no effect of teller gender on description length (girls produced 0.54 [-0.50, 1.55] more words compared to boys), but tellers produced longer descriptions for boy guessers (2.07 [1.01, 3.14]). There were not substantial effects of trial order or interactions between demographics and trial order.

Qualitative observations

Pairs of children varied substantially in what types of descriptions were used, how readily they produced descriptions, and how well they understood and cooperated with the "rules" of the game.

Some children needed reminders from the experimenter about the structure of the game, while others were able to scaffold their own interaction. In one game, the experimenter provided no input after the practice trials because the children rapidly took turns producing referring expressions and selecting the targets. Their advantage could partially be due to being best friends with each other; all children were paired with another child from the same classroom who they were willing to play with, but friendship levels varied.

(earlier × correct + same_speaker × correct|target) + (earlier × correct + same_speaker × correct|expt)

[9] correct∼ trial + centered_speaker_age + speaker_gender_num + centered_listener_age + listener_gender_num + (trial|game) + (trial|target) + (trial|expt)

[10] words∼ trial × (centered_speaker_age + speaker_gender_num + centered_listener_age + listener_gender_num) + (trial|game) + (trial|target) + (trial|expt)

Occasionally, a guesser would provide descriptions of the images ("The walking person or the standing person that's holding something?") so the teller just had to indicate which one rather than self-generate a description.

Some children asked clarifying questions of their partner, but knowing how much information to provide was not always consistent. In one game, child A described the figure as "A human", prompting child B to note "There's two humans." Later in the game, child B used the description "A human", leading child A to ask "Which one?", which child B clarified with "The one that is walking". This anecdote illustrates how emerging abilities can be inconsistent – finding a description inadequate as a guesser does not always translate into providing a more informative description as a teller.

Other pairs of children relied on the experimenter to say whose turn it was or prompt them with options to make a guess or ask questions, and some children struggled to formulate useful questions (asking their partner just "what do you see?"). Even for children who struggled, their behaviors such as gesturing, trying to turn their iPads to show their partner, or asking the researcher for help, are indicative of attempts to solve the communication problem (albeit in ways that did not follow the "rules" of the game).

In instances where children did not communicate successfully, it is difficult to know what the limiting factor was. In general, our qualitative impression was that children were more likely to hit limits related to understanding the game structure (such as when tellers tapped their own iPad), cooperating with the game (such as guessers immediately pressing an option before the teller had a chance to say anything), or thinking the task was about correctly "naming" the shape (saying "I don't know what that's called"), than for children to specifically struggle with how to describe an unknown referent adequately. However, pairs of children also varied significantly in how sensitive they were to collaborating with their partner and adapting descriptions based on their partner's descriptions and their prior success.

Comparison with other iterated reference game datasets

To situate the behavior findings of the children in context, we visually compare the child-child data from our experiments to the most-similar adult-child and adult-adult datasets that use the same stimuli. We use Refbank [version 12.0; Boyce et al. (2026)] which provides harmonized data across a number of iterated reference games. For the experiment 1 and 2 stimuli, we have comparisons from Leung et al. (2024) on 4-year-old and parent dyads and adult-adult dyads, playing face-to-face on 2-option choices. For the experiment 3 stimuli, the most similar available paradigms are from Hawkins et al. (2021) and Hawkins et al. (2023) which are adult-adult dyads typing descriptions to each other online with time limits, and 4-option choice.

The results of 4 metrics available from the Refbank pipeline are shown in Figure 6. The child-child data is not an outlier on any of these metrics. Child dyads are less accurate than adult dyads, but generally similar to the child-adult dyads (Figure 6A). Figure 6B shows length of referential utterances; note that for cross-dataset comparison, this measure includes all words rather than omitting non-referential words and repetitions. Child-child pairs are in between the adult-adult and child-adult pairs on each repetition on the Experiment 1 & 2 stimuli; children say more in Experiment 3 than adults typed. The adults do shorten their

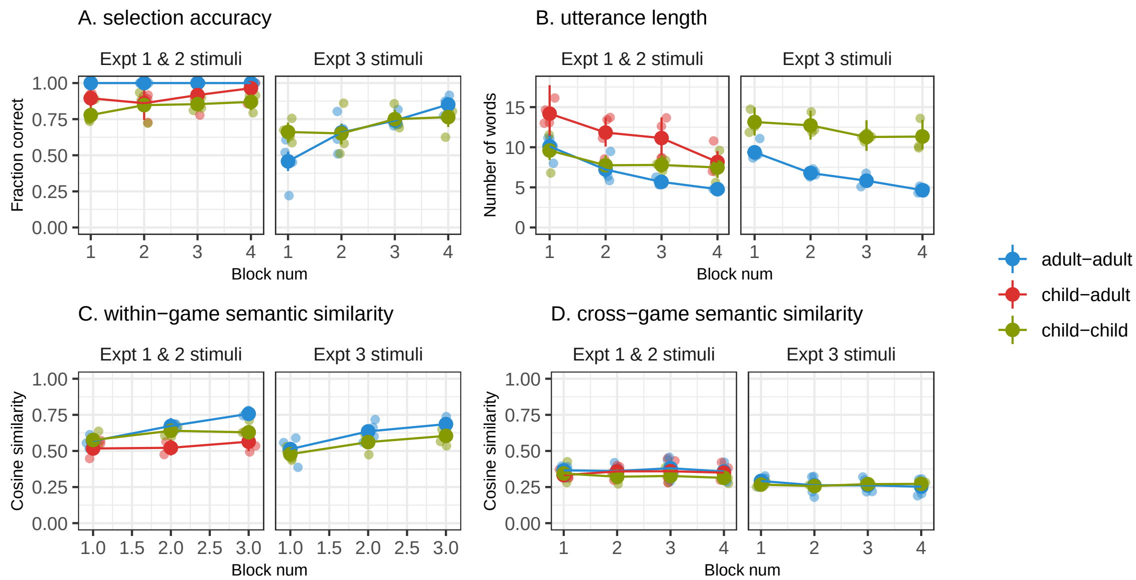


Figure 6. Comparison between this child-child dataset and the most similar other datasets on the same stimuli from Refbank on 4 measures of convention formation.

utterances over repetitions, but much less dramatically than with larger option sets.

Using S-BERT embeddings, Figure 6C shows similarity between a description and the next repetition description to the same target, and Figure 6D shows similarity between descriptions to the same target in the same repetition across games. All conditions show some increase in within-game semantic similarity, with the child-child dyads patterning in between the adults and the child-parent pairs. None of the experiments show much between-game differentiation.

Overall, children show differences of degree compared to adult dyads. Child-child dyads show comparable performance to child-adult dyads, and in some cases are more similar to adult dyads than the child-adult dyads are. While parents certainly scaffold their interactions with children, parental scaffolding is not required to support children's referential abilities at this age.

General discussion

Language is a powerful tool for communication, and adult language users can communicate with each other in a partner- and context-sensitive way. Learning to deploy language effectively to achieve communicative goals is part of what children learn when they learn a language, but how well can young children use language to communicate with each other about images without canonical names? Iterated reference games are a rich tradition for studying the emergence of partner- and context- specific referring expressions in adults. In the current work, we presented children with a scaled down version of an iterated reference game to see how their developing linguistic and pragmatic skills would allow them to communicate and coordinate with each other.

Across 3 experiments and 82 pairs of 4 and 5-year-old children, we tested how well children could produce referential expressions that allowed their partner to find a matching abstract shape. Children varied substantially in what sorts of descriptions they produced, but overall accuracy was high (81%), indicating that children were generally able to produce adequate descriptions. While this task was substantially scaled down relative to measures used for adult competence, it does suggest that the relevant communication skills are present at least in rudimentary form by the end of the preschool years.

Unlike adults, children did not display a shortening of referential expressions over the course of the game. The typical shortening of adult utterances may be contingent on the initial production of longer and over-informative utterances, a condition that does not occur as children produce shorter utterances in this context and are more likely to be under-informative in other referential contexts. Children did show signs of adapting their utterances to their partner and the conversational context. Children's descriptions became increasingly similar to descriptions in the last block and children's successful utterances were more similar to future utterances, suggesting that children adapted their descriptions and coordinated with their partner towards mutually understandable names for each target, even if these names were not getting shorter.

By reducing the task demands and scaffolding children's interaction with each other, we were able to see that many 4-5 year old children in our sample were able to coordinate with each other on informative utterances in a repeated matching game. In a post-hoc analysis, we also observed that within this age range, older children are somewhat more successful than younger children, although there is wide individual variability. We do not claim that this ability emerges at 4 years old, although we suspect that younger children may struggle more with even these reduced task demands, making it difficult to test child-child coordination in young children.

Our study was intended as a proof-of-concept test about whether children at this age could coordinate with each other to refer to novel referents. Still, the generalizability of our results is limited by the target population, the target images, and the task structure. We sampled a convenience population of children at a university nursery school. We used two sets of 4 tangram images, which may be easier to distinguish and have higher codability than other target images used in adult iterated reference games. We specifically targeted children's abilities to construct referring expressions that can be jointly understood, so children were provided scaffolding around taking turns and talking to their partner.

The study of children's pragmatic language use can be approached from different points along the spectrum of experimental control to ecological validity. Iterated reference games have been used as a model system in the adult pragmatics literature because they balance the experimental control over the stimuli with the open-endedness of interactive conversation. Controlled experiments on children's referential communication have established some of the factors that children are sensitive to, and observational work has established that children frequently talk with communicative intent. Use of this model system in the current study provides support for this method as a way of measuring real-time, ecologically valid conversational dynamics of children, and offer a framework for measuring factors that children are sensitive to in their communicative interactions. Experiments that elicit task-specific conversation bridge between these levels with controlled tasks that elicit ecologically valid communicative behavior.

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Data and Code Availability

The experimental set-up, analysis code, de-identified transcripts, and performance data for all experiments is available at https://osf.io/7d3ne/files/github.

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